

```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "deepesh79/trend-nginx-app"
    }

    stages {

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $IMAGE_NAME:latest .'
            }
        }

        stage('Push Image to DockerHub') {
            steps {
                withCredentials([usernamePassword(
                    credentialsId: 'dockerhub-creds',
                    usernameVariable: 'DOCKER_USER',
                    passwordVariable: 'DOCKER_PASS'
                )]) {
                    sh '''
                        echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin
                        docker push $IMAGE_NAME:latest
                    '''
                }
            }
        }

        stage('Deploy to Kubernetes') {
            steps {
                sh 'kubectl apply -f k8s/'
            }
        }
    }
}
```

```
pipeline {
    agent any

    environment {
        IMAGE_NAME = "deepesh79/trend-nginx-app"
    }

    stages {

        stage('Build Docker Image') {
            steps {
                sh 'docker build -t $IMAGE_NAME:latest .'
            }
        }

        stage('Push Image to DockerHub') {
            steps {
```

```
ubuntu@ip-172-31-5-35:~/Trend$ nano Jenkinsfile
ubuntu@ip-172-31-5-35:~/Trend$ sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Hit:4 https://download.docker.com/linux/ubuntu noble InRelease
Get:5 https://apt.releases.hashicorp.com noble InRelease [12.9 kB]
Ign:6 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:7 https://pkg.jenkins.io/debian-stable binary/ Release
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7344 B]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
```

```
ubuntu@ip-172-31-5-35:~/Trend$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
--2025-12-26 11:01:08-- https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 151.101.210.133, 2a04:4e42:31::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|151.101.210.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3175 (3.1K) [application/octet-stream]
```

```
ubuntu@ip-172-31-5-35:~/Trend$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-23 17:25:59 UTC; 2 days ago
 Main PID: 25428 (java)
    Tasks: 57 (limit: 2204)
  Memory: 744.6M (peak: 774.5M)
     CPU: 11min 5.754s
    CGroup: /system.slice/jenkins.service
            └─25428 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Dec 24 06:15:01 ip-172-31-5-35 jenkins[25428]: 2025-12-24 06:15:01.871+0000 [id=3127] INFO o.j.p.g.w.s.DefaultPushGHEventSubs>
Dec 24 06:15:02 ip-172-31-5-35 jenkins[25428]: 2025-12-24 06:15:02.546+0000 [id=3744] INFO c.c.jenkins.GitHubPushTrigger$1#ru>
Dec 24 06:21:55 ip-172-31-5-35 jenkins[25428]: 2025-12-24 06:21:55.628+0000 [id=3219] INFO o.j.p.g.w.s.DefaultPushGHEventSubs>
Dec 24 06:21:55 ip-172-31-5-35 jenkins[25428]: 2025-12-24 06:21:55.629+0000 [id=3219] INFO o.j.p.g.w.s.DefaultPushGHEventSubs>
Dec 24 06:21:56 ip-172-31-5-35 jenkins[25428]: 2025-12-24 06:21:56.178+0000 [id=3854] INFO c.c.jenkins.GitHubPushTrigger$1#ru>
Dec 25 17:25:52 ip-172-31-5-35 jenkins[25428]: 2025-12-25 17:25:52.385+0000 [id=12710] INFO hudson.util.Retrier#start: Attempt>
```

```
ubuntu@ip-172-31-5-35:~/Trend$ git add Jenkinsfile
```

```
ubuntu@ip-172-31-5-35:~/Trend$ git status
```

On branch main

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

new file: Jenkinsfile

Untracked files:

(use "git add <file>..." to include in what will be committed)

aws/

awscli2.zip

```
ubuntu@ip-172-31-5-35:~/Trend$
```

```
ubuntu@ip-172-31-5-35:~/Trend$ git commit -m "Jenkinsfile"
```

Auto packing the repository in background for optimum performance.

See "git help gc" for manual housekeeping.

warning: The last gc run reported the following. Please correct the root cause and remove .git/gc.log

Automatic cleanup will not be performed until the file is removed.

warning: There are too many unreachable loose objects; run 'git prune' to remove them.

```
[main 4c95e71] Jenkinsfile
```

```
Committer: Ubuntu <ubuntu@ip-172-31-5-35.ap-south-1.compute.internal>
```

Your name and email address were configured automatically based

on your username and hostname. Please check that they are accurate.

You can suppress this message by setting them explicitly. Run the following command and follow the instructions in your editor to edit


```
1 file changed, 37 insertions(+)
create mode 100644 Jenkinsfile
ubuntu@ip-172-31-5-35:~/Trend$
```

```
ubuntu@ip-172-31-5-35:~/Trend$ git push origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 2 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (15/15), 3.21 KiB | 3.21 MiB/s, done.
Total 15 (delta 3), reused 11 (delta 1), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/deepeshchandnani/trend-app.git
    16f2b47..4c95e71  main -> main
ubuntu@ip-172-31-5-35:~/Trend$
```

+

New Item

New Item

Enter an item name

trend-ci-cd-pipeline

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK



- ☐ Discard old builds ?
- ☐ Do not allow concurrent builds
- ☐ Do not allow the pipeline to resume if the controller restarts

☒ GitHub project

Project url ?

`https://github.com/deepeshchandnani/trend-app.git`

Advanced ▼



GitHub hook trigger for GITScm polling



Definition

Pipeline script from SCM



SCM ?

Git



Repositories ?

Repository URL ?



`https://github.com/deepeshchandnani/trend-app.git`

Credentials ?

- none -



+ Add

Branches to build ?

Branch Specifier (blank for 'any') ?



*/main

+ Add Branch

Save

Apply



Credentials

Configure credentials

P

Store ↓

Domains



System

(global)

Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
This credential domain is empty. How about adding some credentials?			

Icon:

S

M

L

New credentials

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

deepesh79

☐ Treat username as secret ?

Password ?

.....

ID ?

dockerhub-creds

Description ?

DockerHub credentials

Create



Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
 dockerhub-creds	deepesh79/***** (DockerHub credentials)	Username with password	DockerHub credentials 

Icon:

S

M

L

```
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] sh
+ docker build -t deepesh79/trend-nginx-app:latest .
ERROR: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push Image to DockerHub)
Stage "Push Image to DockerHub" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
Stage "Deploy to Kubernetes" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 1
Finished: FAILURE
```

```
ubuntu@ip-172-31-5-35: ~$ sudo usermod -aG docker jenkins
```

```
ubuntu@ip-172-31-5-35: ~$
```



```
ubuntu@ip-172-31-5-35: ~$ sudo systemctl restart docker
```

```
ubuntu@ip-172-31-5-35: ~$
```

```
ubuntu@ip-172-31-5-35: ~$ sudo systemctl restart jenkins
```

```
ubuntu@ip-172-31-5-35: ~$
```

```
ubuntu@ip-172-31-5-35: ~$ sudo su - jenkins
```

```
docker ps
```

```
+ echo ****
```

```
+ docker login -u deepesh79 --password-stdin
```

WARNING! Your credentials are stored unencrypted in '/var/lib/jenkins/.docker/config.json'.

Configure a credential helper to remove this warning. See

<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
+ docker push deepesh79/trend-nginx-app:latest
```

The push refers to repository [docker.io/deepesh79/trend-nginx-app]

33f95a0f3229: Waiting

0abf9e567266: Waiting

e2c5ba970118: Waiting

```
8c87104c44fc: Waiting
1074353eec0d: Waiting
authentication required - access token has insufficient scopes
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
Stage "Deploy to Kubernetes" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 1
Finished: FAILURE
```

Create access token

A personal access token is similar to a password except you can have many tokens and revoke access to each one at any time. [Learn more](#) 

Access token description

deepesh5

Expiration date

None



Optional

Access permissions

Read, Write, Delete



Read, Write, Delete tokens allow you to manage your repositories.

Cancel

Generate

New credentials

Kind

Username with password

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

Username ?

deepesh79

☐ Treat username as secret ?

Password ?

.....

ID ?

dockerhub-creds

Description ?

dockerhub credentials

Create

Global credentials (unrestricted)

+ Add Credentials

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
 dockerhub-creds	deepesh79/***** (dockerhub credentials)	Username with password	dockerhub credentials 

Icon: S M **L**


```
13b009ca68ba: Pushed
latest: digest: sha256:6a39eaf9bf20f5ecc0d30a8052fa2471fcf49995fd93aec12a38d15634dd7197 size: 856
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] sh
+ kubectl apply -f k8s/
error validating "k8s/deployment.yml": error validating data: failed to download openapi: <html><head><meta http-equiv='refresh' content='1;url=/login?from=%2Fopenapi%2Fv2%3Ftimeout%3D32s'><script id='redirect' data-redirect-url='/login?from=%2Fopenapi%2Fv2%3Ftimeout%3D32s' src='/static/a1d7f1c2/scripts/redirect.js'></script></head><body style='background-color:white; color:white;'>
Authentication required
<!--
-->

</body></html>; if you choose to ignore these errors, turn validation off with --validate=false
error validating "k8s/service.yml": error validating data: failed to download openapi: <html><head><meta http-equiv='refresh' content='1;url=/login?from=%2Fopenapi%2Fv2%3Ftimeout%3D32s'><script id='redirect' data-redirect-url='/login?from=%2Fopenapi%2Fv2%3Ftimeout%3D32s' src='/static/a1d7f1c2/scripts/redirect.js'></script></head><body style='background-color:white; color:white;'>
Authentication required
<!--
-->

</body></html>; if you choose to ignore these errors, turn validation off with --validate=false
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
ERROR: script returned exit code 1
Finished: FAILURE
```

```
jenkins@ip-172-31-5-35:~$ aws eks update-kubeconfig \
  --region ap-south-1 \
  --name trend-cluster
Added new context arn:aws:eks:ap-south-1:284419413658:cluster/trend-cluster to /var/lib/jenkins/.kube/config
jenkins@ip-172-31-5-35:~$
```

```
ubuntu@ip-172-31-5-35:~$ aws eks update-kubeconfig \
  --region ap-south-1 \
  --name trend-cluster
Added new context arn:aws:eks:ap-south-1:284419413658:cluster/trend-cluster to /home/ubuntu/.kube/config
ubuntu@ip-172-31-5-35:~$ □
```

```
ubuntu@ip-172-31-5-35:~$ sudo mkdir -p /var/lib/jenkins/.kube
sudo cp /home/ubuntu/.kube/config /var/lib/jenkins/.kube/config
sudo chown -R jenkins:jenkins /var/lib/jenkins/.kube
ubuntu@ip-172-31-5-35:~$
```



#4 (Dec 26, 2025, 12:24:06 PM)



Add description

Keep this build forever



Started by user [admin](#)

Started 21 sec ago

Took [13 sec](#)



This run spent:

- 11 ms waiting;
- 13 sec build duration;
- 13 sec total from scheduled to completion.



git

Revision: 4c95e713d059f9c7d649d1b8270c6bb20e818329

Repository: <https://github.com/deepeshchandnani/trend-app.git>

- refs/remotes/origin/main



No changes.

```
+ kubectl apply -f k8s/
deployment.apps/trend-app unchanged
service/trend-service unchanged

[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Today



#4

12:24 PM



agent {

environment {

 IMAGE_NAME = "deepesh79/trend-app"

}

stages {


```
ubuntu@ip-172-31-5-35:~/Trend/k8s$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
------	-------	--------	----------	-----

trend-app-b4f86468f-596jh	1/1	Running	0	21h
---------------------------	-----	---------	---	-----

trend-app-b4f86468f-jfc9f	1/1	Running	0	21h
---------------------------	-----	---------	---	-----

trend-app-b4f86468f-172-31-5-35-4m	1/1	Running	0	21h
------------------------------------	-----	---------	---	-----

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
deepesh79/trend-app:latest	1f742c9574aa	99.6MB	32MB	



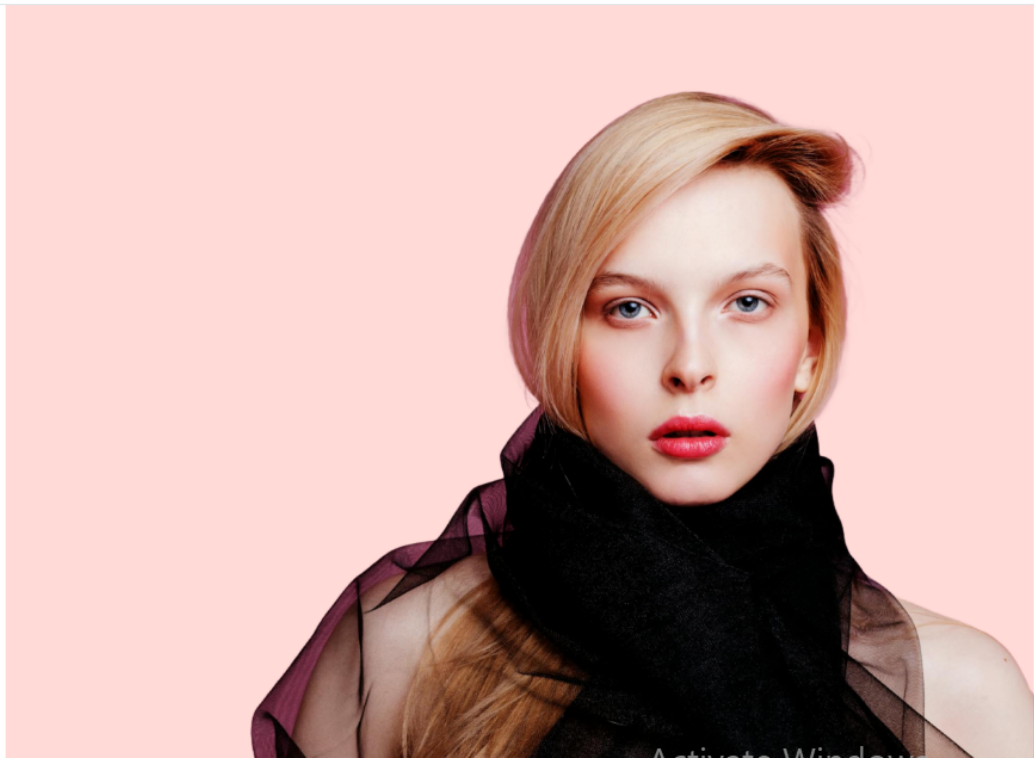
HOME COLLECTION ABOUT CONTACT



— OUR BEST SELLERS

Latest Arrivals

SHOP NOW —



Settings

Recent Deliveries

We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in [our developer documentation](#).

Payload URL *

http://13.201.93.108:8080/github-webhook/

Content type *

application/json

Secret

SSL verification



By default, we verify SSL certificates when delivering payloads.

☒ Enable SSL verification ☐ Disable (not recommended)

Activate Windows

Go to Settings to activate Windows

Which events would you like to trigger this webhook?

- ☒ Just the push event.
 - ☐ Send me **everything**.
 - ☐ Let me select individual events.
-

☒ **Active**

We will deliver event details when this hook is triggered.

Update webhook

Delete webhook

Webhooks / Manage webhook

Settings		Recent Deliveries	
✓	 36ac63da-e273-11f0-8f32-10ace0fa8296 push	2025-12-26 21:24:53	...
✓	 9a139c3c-e24a-11f0-9271-51f77fb5dc75 push	2025-12-26 16:34:09	...
✓	 d7d2d92c-e090-11f0-8010-c04e79c2d75b push	2025-12-24 11:51:55	...
✓	 e1254a60-e08f-11f0-8055-4c6ad351495e push	2025-12-24 11:45:02	...
✓	 d0f19cee-e08e-11f0-9849-2b18e8fc15a7 push	2025-12-24 11:37:25	...
✓	 1a1d1a0c-e084-11f0-99e9-3833a0c7ce3b push	2025-12-24 10:20:43	...
✓	 d786df12-e082-11f0-83fa-c233f7d63aec push	2025-12-24 10:11:42	...
✓	 9eda515e-e081-11f0-98ca-f60ece68f798 ping	2025-12-24 10:02:58	...